**Year 4 Project**
[CSC1097]

# Tag Battle Royale

[Functional Specification]


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# 1. Introduction

## 1.1 Overview

Tag Battle Royale is an app that facilitates the classic game of tag but with a twist to put it on a much larger scale. The game is designed to be played in a large space such as an outdoor park, and split into zones. Each player is shown the designated zones to play in and when those zones will close. Players can not be in a closed zone. The app will use GPS and incorporate features such as a lobby system and map maker to make the game easy to play.

Tag Battle Royale is fast-paced, chaotic, and fun, giving players the thrill of classic tag on a massive scale, built using Flutter for android devices. Tag Battle Royale is designed to be flexible and take consideration of the context it is played in, including no internet connection environments.

## 1.2 Business Context

Tag Battle Royale was originally created by a member of our team and a group of his friends. It was played by using a texting app and drawing on screenshots. The game was very fun and made the group confused as to why there are not more of these types of large-scale social games on the market.

Tag Battle Royale aims to fill this gap and will be suitable for events with friends, team building exercises, schools or parties. It is designed with all ages in mind but with a focus on teens and young adults. Teens and young adults are especially drawn to fast-paced, social, and active games, making them the ideal audience for a large-scale, movement-based experience. They're also highly engaged with mobile technology, which enhances the gameplay and encourages competitive, strategic, and collaborative interactions that keep the game exciting and dynamic.

There are not many similar apps on the market to Tag Battle Royale. There are many apps for offline group games such as *Heads Up* and some that combine multiplayer and GPS features such as *Pokemon Go*, but none that are multiplayer with GPS to be played with a group. The only partially equivalent game is called Tagtile that was discontinued in 2012.

There are a number of ways that could be looked at to monetize Tag Battle Royale:

- **Advertising**
  The app could feature internal banner ads at the edges of the screen, or large advertisements to be skipped could be shown before hosting or joining a game. These ads mustn't interrupt the core gameplay of the app so during the playing phase of the game there can't be much interference for the user.

- **Paid Version**
  A popular model for mobile games and apps is to have a free version downloadable on digital storefronts and then advertise a paid app "upgrade" within the app. This alternate version could provide the user with benefits like no ads or extra features.

  An alternate model of payment is a subscription model. Instead of having one payment that provides the benefits of the paid version forever, have the user pay a smaller amount for a limited time of benefits, such as a month. This model is typically more suitable for apps that the user interacts with regularly over time. Tag Battle Royale is designed to be used occasionally in a burst during a social occasion and because of this, having a subscription model would be detrimental to sales.

## 1.3 Glossary

- **Peer-to-Peer**
  Network architecture where participants or *peers* act both as server and client. Unlike a traditional server-client model, no central server is needed to act as an authority to store and serve data.
- **Scatternet**
  A scatternet is a wireless ad hoc network formed by connecting two or more Bluetooth networks together referred to as piconets, allowing for extended communication beyond the range of a single piconet and for the number of devices able to communicate with each other through Bluetooth, to be extended. [1]
- **Lobby**
  Traditionally a virtual space where participants of a game gather before it starts and where rules and roles can be set.
- **Map/Play Space/Zone**
  Before a game of Tag Battle Royale starts, a map and play space is chosen that participants are not allowed to leave during play. This space is broken up into zones that are closed over the course of the game, further limiting the play space in which participants can stay.
- **Chaser/Runner**
  Players in a game of Tag Battle Royale are designated the role of Chaser or Runner. The Chasers' objective is to find and place a hand on, also known as *tag*, the Runners, which converts the Runner's role to a Chaser. Once all Runners are tagged, the game ends.

# 2. General Description

## 2.1 Product / System Functions

Below is a list of the main functions of the project. This is a preliminary list and is open for changes should anything is thought of needing to be changed during the implementation stage of development. These will be expanded upon in Section 3.

- **Home / Lobby**
  Tag Battle Royale provides a home page for users to navigate to the various features of the application as seen in figure 1. Many features such as the replay viewer and the play phase of the game will be interacted with through a lobby template shown in figure 2.

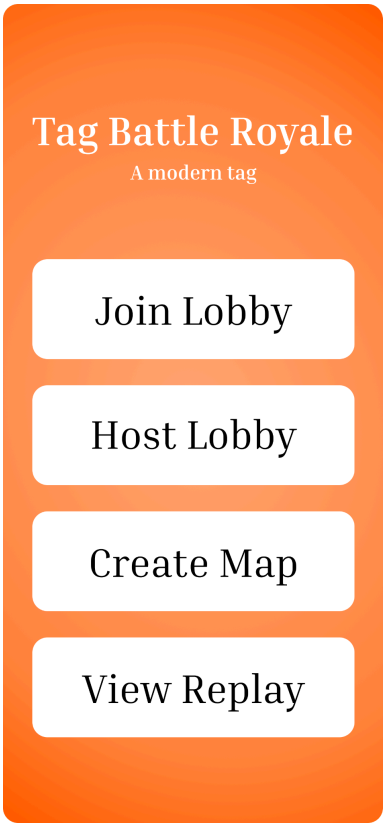



*Figure 1. Home Page*

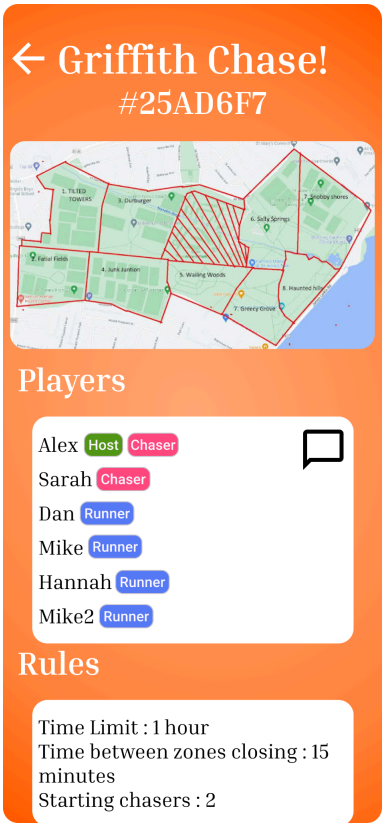



*Figure 2. Lobby Page*

- **Offline/Online Modes**
  Tag Battle Royale will mostly be a game played outdoors. It is not expected that players will have a consistent internet connection for each game, therefore two separate modes are planned to be developed that the host can choose between before creating a lobby. The offline mode will need players to connect to a lobby only once before the game begins using a Bluetooth peer-to-peer connection, no internet connection required. In the offline mode, the players who begin as Chasers will predefine the order in which zones will close before the game begins. The Runners' devices will download this

information but keep the order hidden until the zone is set to close. Chasers and Runners can view the play space and closed zones as well as be shown their location using satellite GPS, but can't see the location of other players due to connection limitations.

The online mode will allow Chasers to choose which zone to close on the fly and have positions of players displayed on the map using GPS. Chasers will be able to see the location of other Chasers and Runners that are in closed zones or out of play space bounds. Online mode is expected to be played when each player has a cellular data connection or on a college campus with widespread internet access.

- **Host/Join a Lobby**
  A user can host a lobby. If online, this will generate a code that can be shared to other users and is used to join the lobby. The host that creates the lobby can adjust settings such as time between zone closures and select the space to be played in. Players will be able to view the map, type messages, and designate their roles between each other as Chaser or Runner.

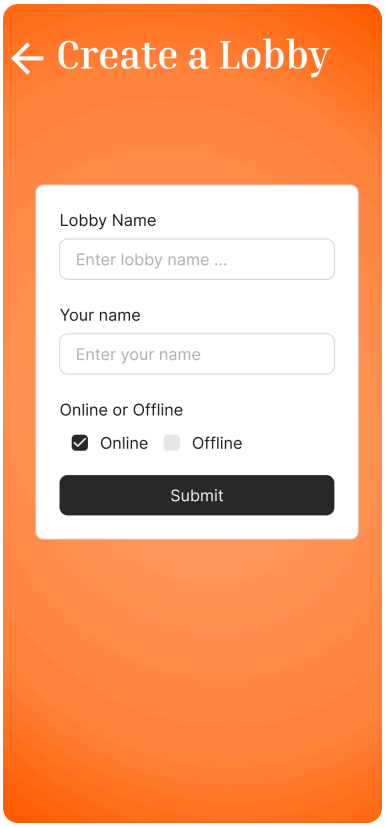



*Figure 3. Host Lobby*

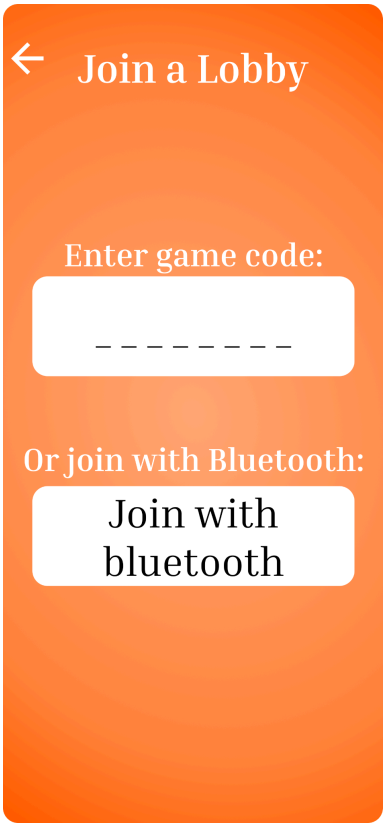



*Figure 4. Join Lobby*

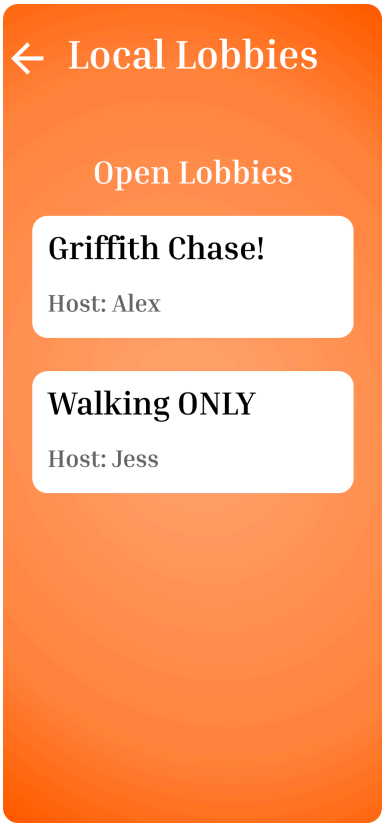



*Figure 5. Local Lobbies*

- **Map Creator**
  Using an embedded map, a user can select a space to play in, e.g a park, and create a map. They can designate the zones on the selected space by drawing them into the selected space, an example of how this will look is shown in figure 6. This map can then be selected when creating a lobby.

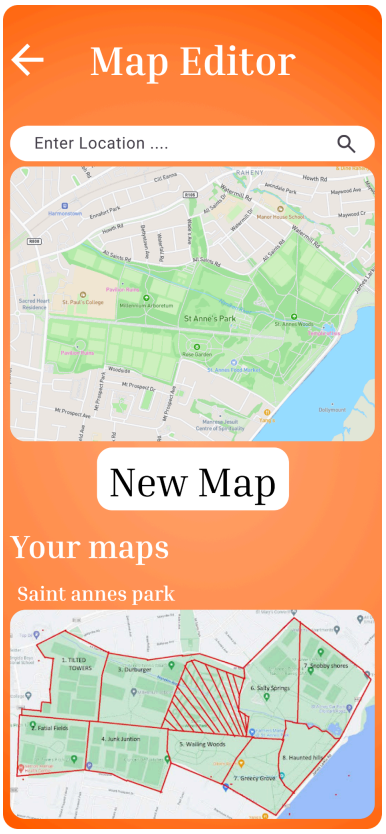



*Figure 6. Map Editor*




*Figure 7. Example Map*

- **Replay/History**
  Information about the game played will be saved after each game. The map played in and the location of each player during the game will be used to create a replay. This feature will be automatic in online mode with data being uploaded to the server throughout the game and being coalesced when the game concludes. Local individual player replays can be saved when offline.

## 2.2 User Characteristics and Objectives

The app will be accessible with a user-friendly interface so as to accommodate users of all ages planning to play the game. The primary demographic for the app will be teens and young adults, which are expected to have proficiency with mobile devices. The setup of a game of Tag Battle Royale must be smooth but also provide plenty of options.

Cultivating a good community of users in the aforementioned age range who are capable and excited to tinker with the game setup options, such as the order and frequency of zone closures and different modes, would be very beneficial to the health of the app but the app should be open and accessible enough to users outside the goal age range too. The app will be compatible with Android devices.

## 2.3 Operational Scenarios

Listed below are scenarios in which different users will use the app:

- **Host Game**
  The user who wants to host will open the app and be presented with options to host a game either online or offline. If online mode is chosen, the host will be given a game code which will act as an address and password to their lobby that they can give to other players. If offline is chosen, other players can join through Bluetooth connectivity.

  Afterwards they are taken to the game lobby. The lobby page will allow the host to configure the game.

  - ***Select Map:***
    The host needs to choose a play space in the map tab. The host can choose a previously created map or create a new map. If the host created a new map, they will be brought to the Map Creator page.
  - ***Choose Player Roles:***
    The host will always start as a Chaser. The host can designate players to be Chasers in the players tab, which will show the list of players in the lobby.
  - ***Chat in Game Lobby:***
    The host can switch to the chat tab and send messages in a group chat with the rest of the players in the lobby.
  - ***Setup Game Rules:***
    The host can switch to the game rules tab where they can configure various rules such as time between closures, if order of zone closures are pre-determined or voted on during game (only in online mode) and number of zone closures before game end state.

- ***Start Game:***
  The host chooses when to start the game.

- **Join Game**
  The users who want to join will open the app and be presented with options to join a game either online or offline. If online mode is chosen, the player will need to input a game code given by the host to join their game. If offline is chosen, the player can join through Bluetooth connectivity. Afterwards they are taken to the game lobby. The game lobby page will consist of tabs they will switch through.

  - ***View Map:***
    The player can switch to the map tab to view the play space and zones.
  - ***Chat in Game Lobby:***
    The player can switch to the chat tab and send messages in a group chat with the rest of the players in the lobby.

- **Play Game**
  Once the host starts the game, the game will begin where the host and all players will be brought to the game page.

  - ***Check Map/GPS Functionality:***
    Players can check the map to see the play space as well as all open and closed zones. Players can see their locations on the map, if online mode is chosen, Runners that are outside of the play space or in a closed zone will be visible to Chasers through GPS functionality.
  - ***Chat in Game Lobby:***
    Players can chat in the game lobby if online mode is chosen. Separate chat rooms for Chasers and Runners are available.
  - ***Switch to Chaser (if Runner):***
    When Runners are caught they must open the app, select the 'Caught' button and which player caught them. This will convert their role to Chaser and they'll gain all benefits of Chaser including voting on zone closures if enabled and seeing players outside of permitted areas.
  - ***Vote on Zone Closures (if Chaser and Online):***
    If enabled and the player is designated Chaser, the player can vote on the order in which zones are closed on the fly in online mode. If no votes are cast, a zone is chosen randomly.
  - ***End Game:***
    When certain criteria are met that are set at the start of a game, such as the last zone being closed or a certain number of players being tagged (only available online), the game ends. Games may be manually ended as well.

- **Save and View Replays**
  When a game ends, a replay of the game is saved to all players' devices if playing online. In offline mode, an individual replay of a player's actions is saved to their device. Users can view previous replays which includes showing when players have been caught and their locations on the map.

## 2.4 Constraints

- **Online Mode Server and Security**
  The app's servers must handle concurrent lobbies and be reliable enough to transmit live GPS data. Security is a large concern due to potentially sensitive location data. A security technique such as websockets should be used in the app's servers.

- **Offline Mode Bluetooth Connection**
  The app must be able to transmit data through Bluetooth at the start of a game. Creating a lobby through Bluetooth connectivity is a challenge as an android device typically can not support connecting to many other devices at the same time. A certain network topology for a device to communicate with multiple others will be implemented, where some devices will act as pseudo-hosts to transmit data from multiple devices to the true host. This method of Bluetooth networks is referred to as scatternet. Simply connecting and disconnecting each device to the host one at a time at the beginning of a game is also an option if the previous solution is unable to function.

- **Cheating**
  There is incentive for a Runner to lie and claim that they were not tagged by a Chaser. Video games may have stricter rules to better restrict the possibility of cheating, but Tag Battle Royale is a party game with a certain level of trust expected between players of the group.

  There is not a lot that can be done about cheating in a normal game of Tag or Tag Battle Royale, but one way the app tries to combat this is with its replay feature. Players will need to manually select when they were caught and by whom. Their locations will also be tracked on the map. So the validity of Runner's claims can be tested after a game is over. Additionally, if in online mode, the host has the ability to manually convert a Runner's role at any time.

- **App Usability**
  The app needs to be quick, reliable and easily accessible on the fly. Certain features such as play space and zone selection need to be easy to use. Drawing using a touch screen can be a frustrating prospect due to inaccuracy.

# 3. Functional Requirements

### 3.0.01 Host Game - Offline

**Description:** This function enables a user to host a game of Tag Battle Royale offline. This function would create a "lobby" on the host's client device that would act like a virtual room and server for players to join directly before beginning the game. The host automatically joins the lobby and others can join through Bluetooth connectivity. While in the lobby, the host can configure settings for the upcoming game and players can interact with one another.

**Criticality: Critical to Offline App Functionality**

This function is how a game of Tag Battle Royale is created, it is ranked higher than online because most games of Tag Battle Royale will be played offline due to the nature of the game.

**Technical Issues:**

- Developing the model to host a "server" on a client and having several players connected to it simultaneously will be a large hurdle to overcome. A network topology between devices called a scatternet will need to be implemented, due to Bluetooth connectivity usually being unable to support several devices being linked together at once.
- Connecting one at a time at the beginning of a game to upload and download data between clients is an option if the other method is unable to be implemented.

**Dependencies:** None.

### 3.0.02 Host Game - Online

**Description:** This function enables a user to host a game of Tag Battle Royale online through an internet connection. This function would create a "lobby" on the game server that would act like a virtual room for players to join before beginning the game and stay in during the game. The host automatically joins the lobby and is provided with a game code that would act as an address and password for other players to join the lobby through an internet connection. While in the lobby, the host can configure settings for the upcoming game and players can interact with one another.

**Criticality: Critical to Online App Functionality**

This function is how a game of Tag Battle Royale is created for online play.

**Technical Issues:**

- An online server needs to be created using a hosting site and be able to host multiple games concurrently. We are planning to achieve this through Serverpod.
- The development team needs to be vigilant about security due to the handling of realtime location data.

**Dependencies:** None.

### 3.0.03 Join Game - Offline

**Description:** This function enables a user to join a game of Tag Battle Royale offline through Bluetooth connectivity. The user connects to the host's offline lobby and will be a part of the game when it begins. The user will be a part of the scatternet and can make use of the lobby features. Without this function, the app won't function offline.

**Criticality: Critical to Offline App Functionality**

This function is how a player joins a game of Tag Battle Royale offline.

**Technical Issues:**

- Same as Host Game - Offline, having multiple players connected to each other offline will be a challenge as a topology needs to be developed for Bluetooth connectivity.

**Dependencies:** 3.0.01 Host Game - Offline

### 3.0.04 Join Game - Online

**Description:** This function enables a user to join a game of Tag Battle Royale online through an internet connection. The user connects to the host's lobby on the backend server and will be a part of the game when it begins. The user can make use of the lobby features.

**Criticality: Critical to Online App Functionality**

This function is how a player joins a game of Tag Battle Royale online. Without this function, the app won't function online.

**Technical Issues:**

- Same as Host Game - Online, multiple players connected to each other online will be a challenge as a server needs to be created that can host multiple games simultaneously. There are security concerns too.

**Dependencies:** 3.0.02 Host Game - Online

### 3.0.05 Create Map

**Description:** This function enables the user to select the location and a play space the game will take place in and its closure zones. The play space is where the players will be confined to. The zones will close on a timer over the course of a game, Runners must exit zones before they close. Selection of play space and zones is done through drawing on a map with the use of the Google Maps library for flutter.

**Criticality: Critical to App Functionality**

This function is how the play space and closure zones are defined.

**Technical Issues:**

- Google Maps needs to be implemented for a fully functioning map and a way to search it.
- Drawing using touch screen needs to be intuitive and shapes need to be recognised.

**Dependencies:** None.

### 3.0.06 Start Game

**Description:** This function starts the game and brings all the players in the lobby from the setup page to the game page. The host decides when this function is called. This function brings all other functions together. Players can view the map, which shows the play space and closure zones, at any time during the course of a game. Runners will be notified when a zone is closing while Chasers will know what zones are closing ahead of time. In online mode, Runners not in open zones will be revealed to Chasers on the map with GPS functionality.

**Criticality: Critical to App Functionality**

This function describes how the game is played.

**Technical Issues:**

- Learning how to properly display the map in conjunction with GPS

**Dependencies:** 3.0.05 Choose Play Space and Zones

### 3.0.07 End Game

**Description:** This function ends the game and brings all players to the home page. The criteria for ending the game is set by the host before the game begins. Common criteria may include all zones getting closed or no Runners remaining. In offline mode, if no Runners are remaining for example, players will need to manually end the game themselves using the 'exit game' button. In online mode, this detection is automatic and the game will end itself. After the game is over, if in online mode, a replay including all players locations throughout the game, when a player switches roles and the state of the play space and zones can be saved to the players' device. If in offline mode, a player can review an individual replay of their own actions after a game.

**Criticality: Not Critical to App Functionality, Very High Importance**

This function describes how the game is ended. This app can function without it but would instill a clunky experience to the user.

**Technical Issues:**

- None.

**Dependencies:** 3.0.06 Start Game

### 3.0.08 Allocate Roles to Users

**Description:** This function allows users to be allocated either the role of Chaser and Runner. A host can make use of this function at the start of a game to set roles. Hosts can set roles during the game too. During the game, Runners must change their role to Chaser when tagged, using the 'Caught' button on the game page. After selecting the button, the player chooses who caught them for replay purposes (an unknown option is available).

**Criticality: Not Critical to App Functionality, Very High Importance**

This function is how roles are allocated. Technically a version of the game without roles can be developed, where the app just displays zone closures.

**Technical Issues:**

- None.

**Dependencies:** None.

### 3.0.09 GPS Functionality

**Description:** This function will be used in conjunction with the map to show player locations. Every player will be able to see themselves on the map in online mode, Runners not in open zones will be revealed to Chasers on the map. In online mode, the Geolocator library will be used to track players with an internet connection. In offline mode, Geolocator will be used to track players with satellite data. Players' location throughout the game will be recorded and saved to the replay.

**Criticality: Not Critical to App Functionality, High Importance**

This function is how the app will track players. Similar to 3.0.07 Allocate Roles to Users, technically a version of this app without GPS functionality can be developed.

**Technical Issues:**

- Learning to use Geolocator will be a challenge.

**Dependencies:** 3.0.05 Choose Play Space and Zones

### 3.0.10 View Replays

**Description:** This function allows users to view replays of their previous games. Replays include all players locations throughout the game, when a player switches roles and the state of the play space and zones can be saved to the players' app.

**Criticality: Not Critical to App Functionality, Medium Importance**

This function allows users to view replays. This is a useful feature but is not critical for the functionality of the game.

**Technical Issues:**

- Players must reconnect with one another after the end of a game. The upload and download of data must be quick and must account for potentially missing data.

**Dependencies:** 3.0.07 End Game

### 3.0.11 Vote on Zone Closures

**Description:** This function allows Chasers to vote on the order in which zones close. In offline mode, the order of zone closures can be random or predetermined by the host before the game. In online mode, votes can be held during the game to decide on the next zone closure. A Chaser will be able to choose what zone they think should be closed by selecting it on the map during a game. This function should create some strategy in the game.

**Criticality: Not Critical to App Functionality, Medium Importance**

This function controls zone closure votes. The host can manually decide the closure of zones at the start of a game or leave it to be random regardless of mode.

**Technical Issues:**

- None,

**Dependencies:** 3.0.05 Choose Play Space and Zones

### 3.0.12 Chat Functionality

**Description:** This function allows players in a lobby to chat to one another. On the setup page players can use this function to message each other in one chat room. In online mode, separate chat rooms can be allocated for Chasers and Runners during a game on the game page.

**Criticality: Not Critical to App Functionality, Low Importance**

This function creates chat rooms. This function is an extra that is not too important for the game as players should be able to easily use external apps for this feature.

**Technical Issues:**

- None.

**Dependencies:** (Optionally) 3.0.07 Allocate Roles to Users

# 4. System Architecture

The following, Figure 8 System Architecture, shows the intended system architecture for Tag Battle Royale. The system acts fundamentally differently in online mode when compared to offline mode, therefore two diagrams have been provided.

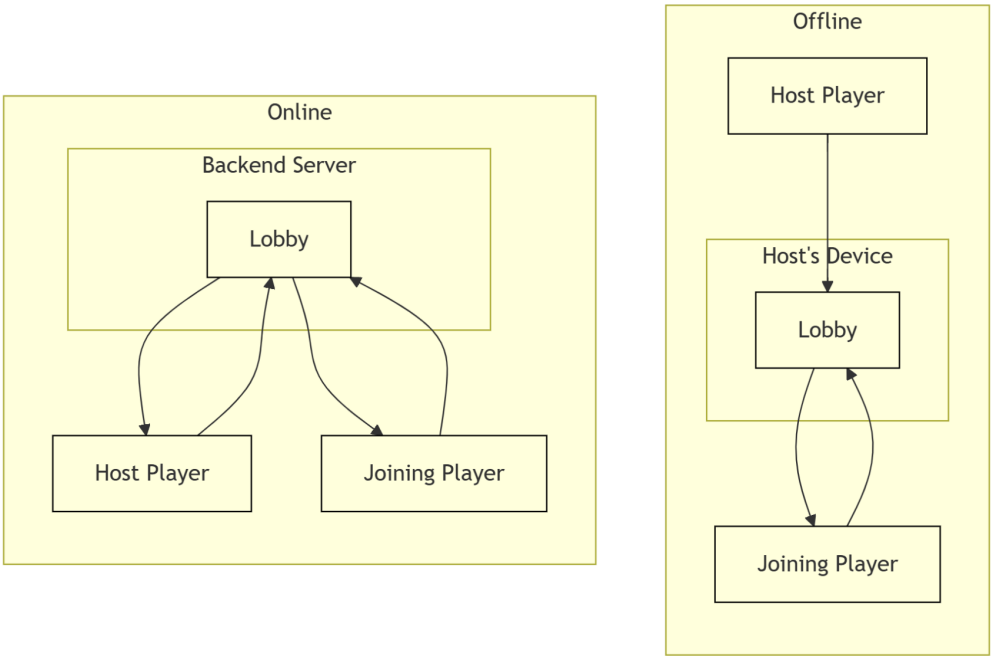



*Figure 8. System Architecture*

### 4.1 Online

For the online mode, the architecture is more traditional. The hosting player creates a lobby on the backend server and other players can join this lobby by connecting to the server. This mode requires an internet connection.

### 4.2 Offline

For the offline mode the system will utilise a Bluetooth peer-to-peer connection. This is to allow players' devices to communicate with each other without an internet connection. The hosting player will host the lobby locally on their device and the other players will connect to the host's device in a network of devices called a scatternet. This is because Bluetooth devices have limitations in the number of active connections they can make.

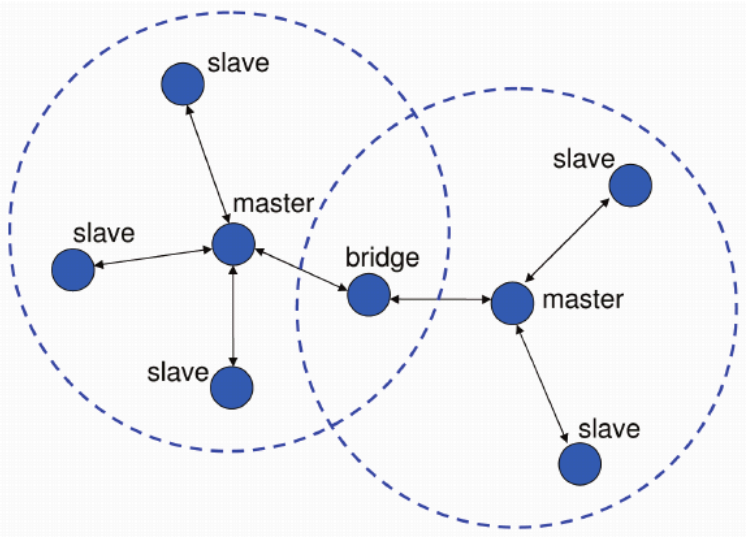



*Figure 9. Scatternet Example [2]*

Figure 9 shows an example of a scatternet in action. In this diagram, using the context of our project, the masters act as pseudo-hosts to the real host. If the number of active Bluetooth connections a master can support is seven, then in this example, fourteen devices can be connected to the host. This version of the game will have reduced functionality due to lack of real-time information access during play.

# 5. High-Level Design

## 5.1 System Flowchart

The following flowchart shows all interaction within Tag Battle Royale:

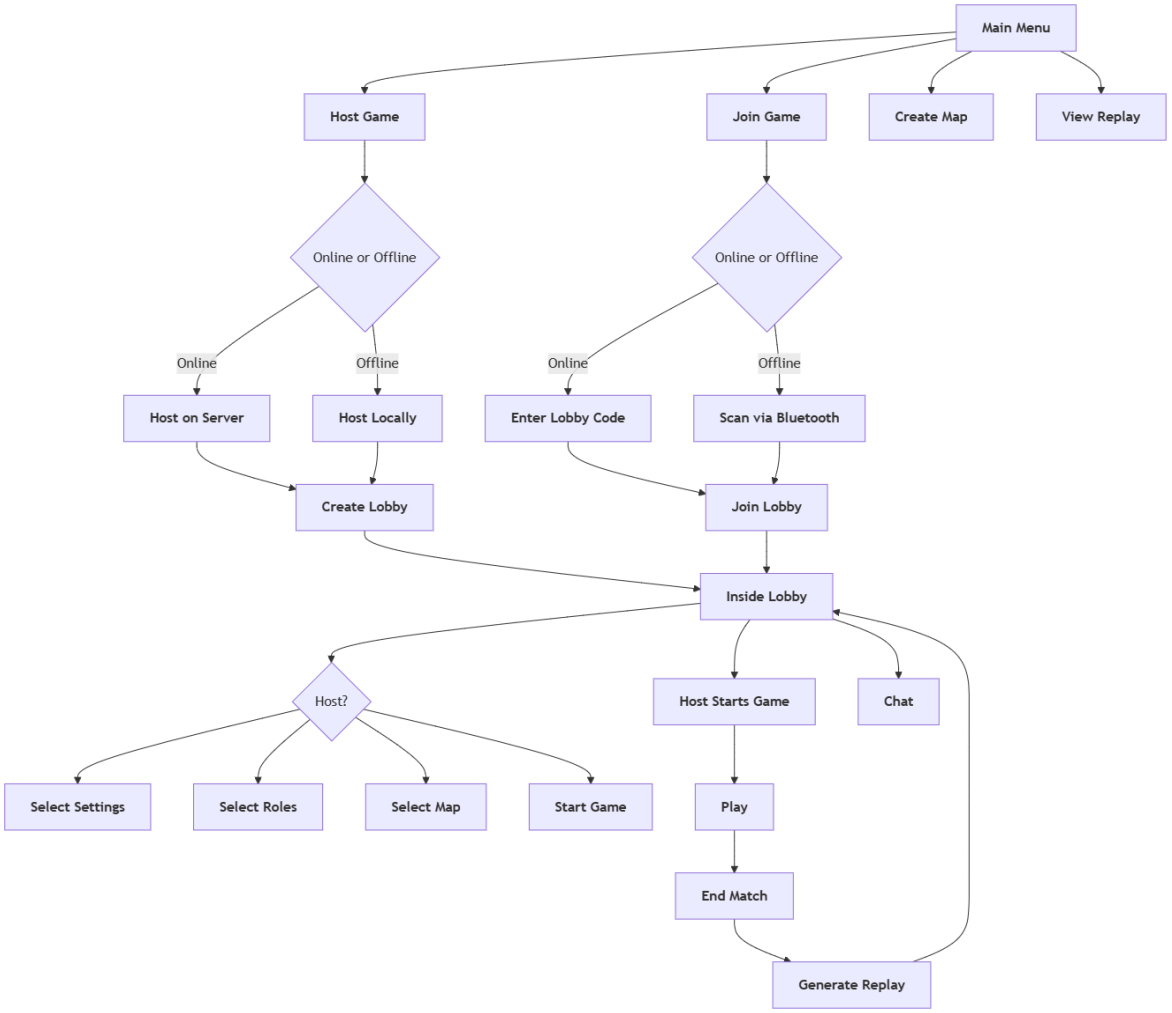



*Figure 10. System Flowchart*

When creating an online game, the hosting player makes a request to the backend server to create a lobby. The server creates and connects the player to this lobby and provides a lobby code. Other players can join this lobby by providing the server with the same lobby code. Tag Battle Royale is played in person so sharing this code will not be an issue. When creating an offline game, the hosting player will host the lobby locally on their device and broadcast the lobby via Bluetooth. Other players can join via Bluetooth.

Once inside a lobby, the host can adjust the settings, roles and map of the game. When entering the map creator, the host will be presented with any maps they have stored along with a map embed. The map embed can be used to create a new map. First it is searched with a search bar or manually, when the host has found their desired location the host can lock the map and an image will be produced. The host

will then draw the boundaries of the map along with the playable zones. The zones are indexed and can be named. The host can save this information along with a name as a new map. This is stored locally.

The host can then choose when to start the game. After the game is finished, a replay is generated. During the play phase of a game, each player's GPS location will be stored along with time it was captured. At the end of a game, each player's location on the map throughout the game will be compiled into replay for the players to watch. If online, the replay of each player can be compiled into a singular replay on the host's device. Replays can be stored locally.

## 5.2 Play Phase State Diagram

The following state diagram shows the states of the play phase for Tag Battle Royale:

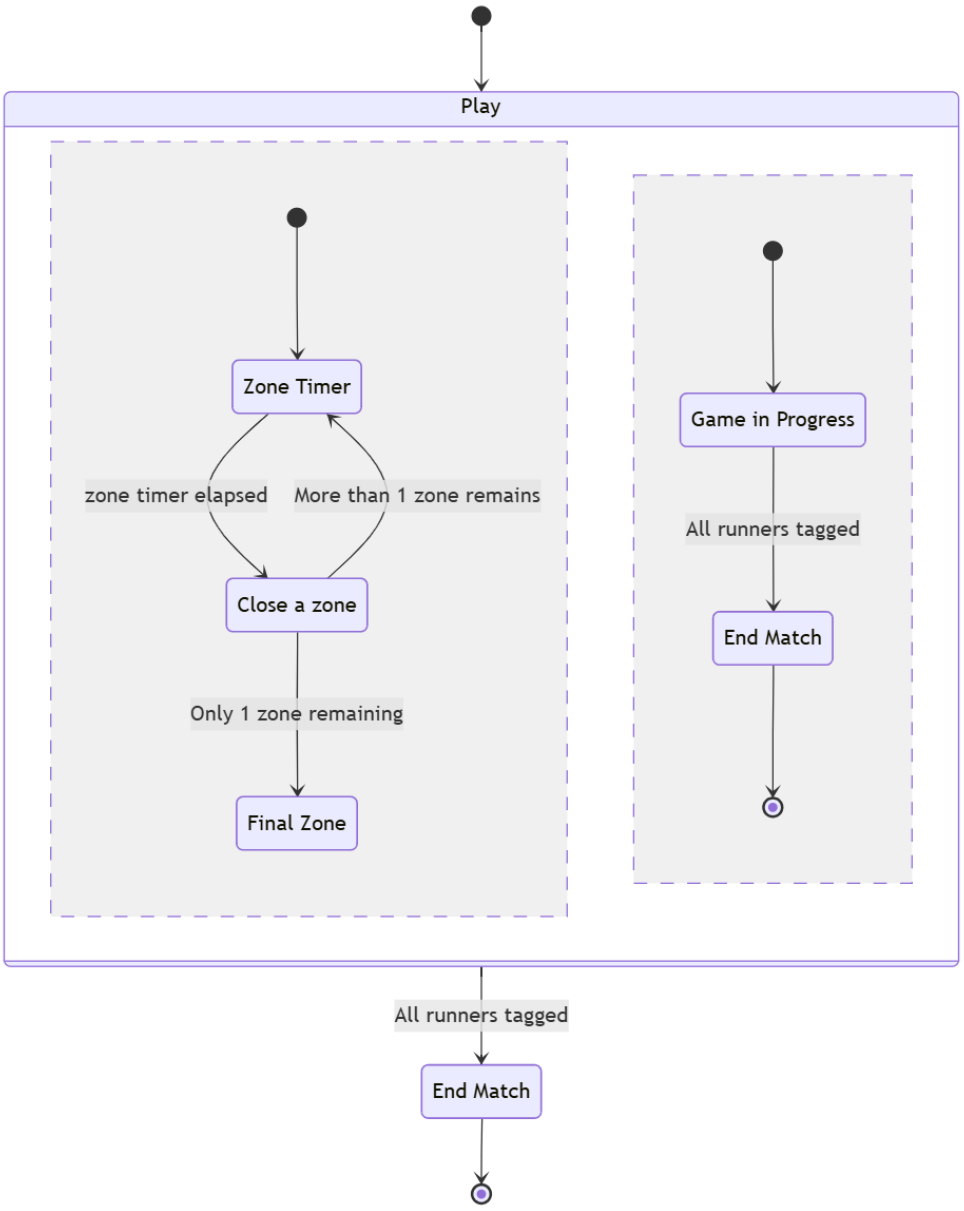



*Figure 11. Play Phase State Diagram*

The online play phase seen in figure 11 consists of the players playing the game and a timer. The chasers try to tag the runners and if all players are turned into chasers then the game is over. The timer starts when the game starts and will notify the chasers every 15 minutes by default. Upon notification, the chasers can close a zone on the map, reducing the area in which the runners can stay. When a zone is chosen, the server will update each player's game, closing that zone. The chasers will have access to a chat for communication and their GPS information will be displayed on the map.

The offline play phase is similar to the online version. Instead of the chasers using their internet connection to dynamically close zones, the zones are pre-chosen and this order is downloaded to each player's phone. When the timer elapses the next zone in the order will close. Only currently closed zones will be visible to the runners but the zone closing order will be visible to the chasers.

## 5.3 Class Diagram

The following diagrams shows all classes within Tag Battle Royale:

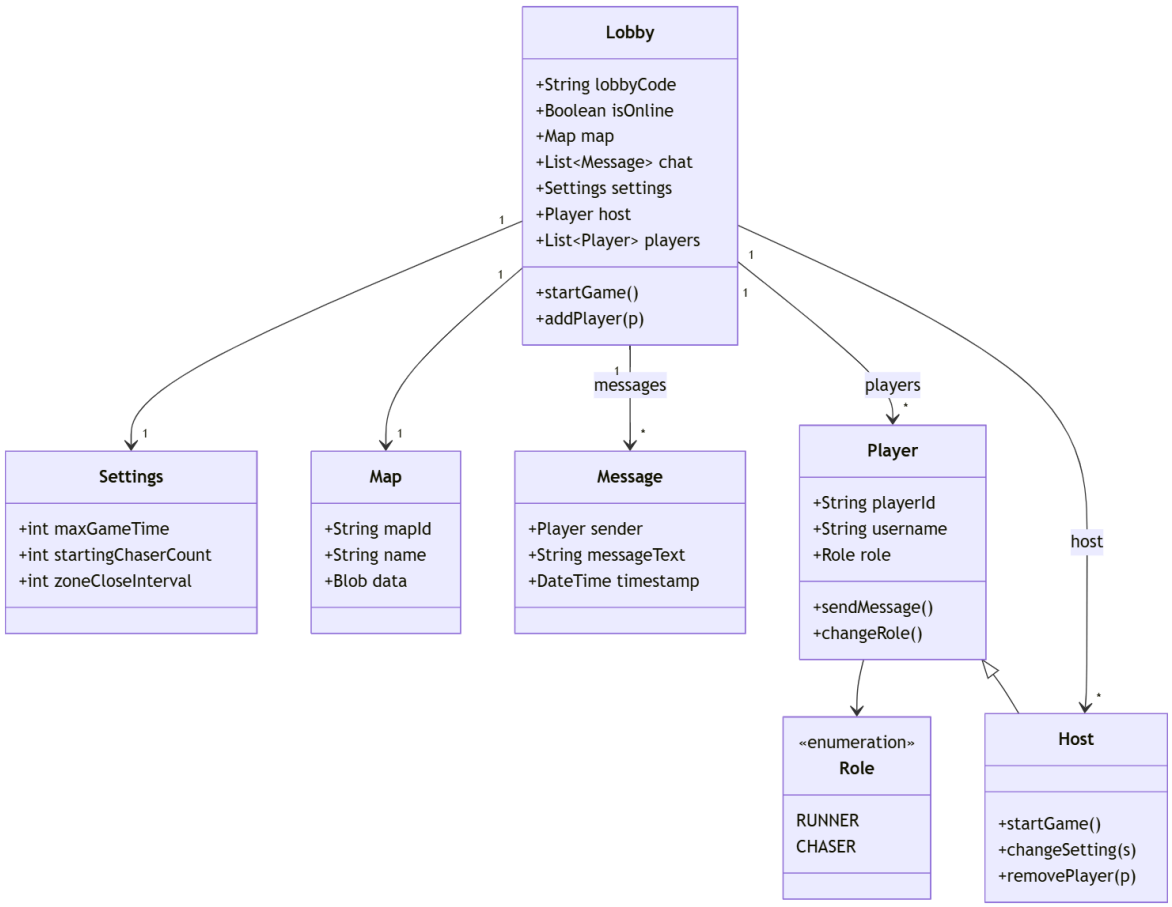
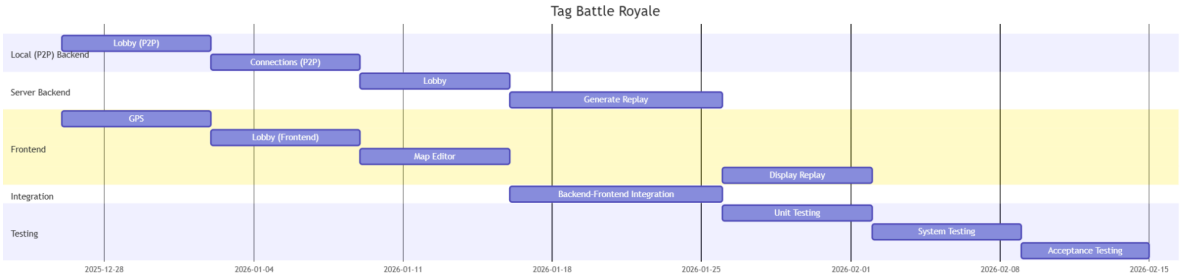



*Figure 12. Lobby Class Diagram*

# 6. Preliminary Schedule




This schedule prioritises development of dependencies before dependent features. Core functions such as the lobby are prioritised with an offline version of the game being focused on first. Time allocation for development is optimistic to allow for an agile approach.

# 7. References

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**Additional Resources:**

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